



# DW01\_incdec

## Incrementer-Decrementer

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

### Features and Benefits

- Parameterized word length

### Revision History

### Description

DW01\_incdec is an incrementer-decrementer. DW01\_incdec either adds 1 to ( $INC\_DEC = 0$ ) or subtracts 1 ( $INC\_DEC = 1$ ) from an input number  $A$  to produce the output  $SUM$ .

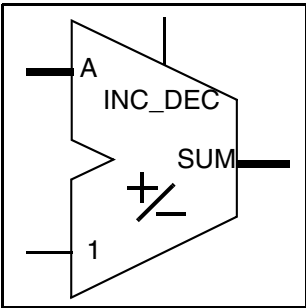


Table 1-1 Pin Description

| Pin Name | Width             | Direction | Function                                                                                                                                            |
|----------|-------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| A        | <i>width</i> bits | Input     | Input data                                                                                                                                          |
| INC_DEC  | 1 bit             | Input     | Increment control <ul style="list-style-type: none"><li>■ 0 = Increment (<math>A + 1</math>)</li><li>■ 1 = Decrement (<math>A - 1</math>)</li></ul> |
| SUM      | <i>width</i> bits | Output    | Increment ( $A + 1$ ) or decrement ( $A - 1$ )                                                                                                      |

Table 1-2 Parameter Description

| Parameter | Values   | Function                     |
|-----------|----------|------------------------------|
| width     | $\geq 1$ | Word length of $A$ and $SUM$ |

Table 1-3 Synthesis Implementations<sup>a</sup>

| Implementation Name | Function                                                                                           | License Feature Required |
|---------------------|----------------------------------------------------------------------------------------------------|--------------------------|
| rpl                 | Ripple carry synthesis model                                                                       | none                     |
| cla                 | Carry look-ahead synthesis model                                                                   | none                     |
| pparch              | Delay-optimized flexible parallel-prefix                                                           | DesignWare               |
| apparch             | Area-optimized flexible architecture that can be optimized for area, for speed, or for area, speed | DesignWare               |

- a. During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the `set_implementation` command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

**Table 1-4     Simulation Models**

| Model                           | Function                             |
|---------------------------------|--------------------------------------|
| DW01.DW01_INCDEC_CFG_SIM        | Design unit name for VHDL simulation |
| dw/dw01/src/DW01_incdec_sim.vhd | VHDL simulation model source code    |
| dw/sim_ver/DW01_incdec.v        | Verilog simulation model source code |

## Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

## HDL Usage Through Operator Inferencing - VHDL

```
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;

entity DW01_incdec_oper is
    generic(wordlength: integer := 8 );
    port(in1      : in STD_LOGIC_VECTOR(wordlength-1 downto 0);
         inc_dec  : in STD_LOGIC;
         sum      : out STD_LOGIC_VECTOR(wordlength-1 downto 0) );
end DW01_incdec_oper;

architecture oper of DW01_incdec_oper is
    signal in_signed,sum_signed: SIGNED(wordlength-1 downto 0);
begin

    in_signed <= SIGNED(in1);
    process (in_signed, inc_dec)
    begin
        if (inc_dec = '1') then
            sum_signed <= in_signed - 1;
        else
            sum_signed <= in_signed + 1;
        end if;
    end process;
    sum <= STD_LOGIC_VECTOR(sum_signed);
end oper;
```

## HDL Usage Through Operator Inferencing - Verilog

```
module DW01_incdec_oper(in1,inc_dec,sum);  
    parameter wordlength = 8;  
  
    input [wordlength-1:0] in1;  
    input inc_dec;  
    output [wordlength-1:0] sum;  
    reg [wordlength-1:0] sum;  
  
    always @(in1 or inc_dec)  
    begin  
        if (inc_dec == 1)  
            sum = in1 - 1;  
        else  
            sum = in1 + 1;  
        end  
    endmodule
```

## HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW01_incdec_inst is
  generic ( inst_width : NATURAL := 8 );
  port ( inst_A          : in std_logic_vector(inst_width-1 downto 0);
        inst_INC_DEC    : in std_logic;
        SUM_inst        : out std_logic_vector(inst_width-1 downto 0) );
end DW01_incdec_inst;

architecture inst of DW01_incdec_inst is
begin
  -- Instance of DW01_incdec
  U1 : DW01_incdec
    generic map ( width => inst_width )
    port map ( A => inst_A, INC_DEC => inst_INC_DEC, SUM => SUM_inst );
end inst;

-- pragma translate_off
configuration DW01_incdec_inst_cfg_inst of DW01_incdec_inst is
  for inst
  end for; -- inst
end DW01_incdec_inst_cfg_inst;
-- pragma translate_on
```

## HDL Usage Through Component Instantiation - Verilog

```
module DW01_incdec_inst( inst_A, inst_INC_DEC, SUM_inst );

    parameter width = 8;

    input [width-1 : 0] inst_A;
    input inst_INC_DEC;
    output [width-1 : 0] SUM_inst;

    // Instance of DW01_incdec
    DW01_incdec #(width)
        U1 ( .A(inst_A), .INC_DEC(inst_INC_DEC), .SUM(SUM_inst) );
endmodule
```

## Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

| Date           | Release       | Updates                                                                                                                                                                        |
|----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2024 | DWBB_202409.0 | ■ Updated date and version; no technical changes                                                                                                                               |
| December 2023  | DWBB_202312.0 | ■ Updated date and version; no technical changes                                                                                                                               |
| March 2019     | DWBB_201903.0 | ■ Removed Table 1-4, “Obsolete Synthesis Implementations”                                                                                                                      |
| January 2019   | DWBB_201806.5 | ■ Updated example in “ <a href="#">HDL Usage Through Component Instantiation - VHDL</a> ” on page 5<br>■ Added this Revision History table and the document links on this page |

## Copyright Notice and Proprietary Information

© 2024 Synopsys, Inc. All rights reserved. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

### Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

### Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

### Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>.

All other product or company names may be trademarks of their respective owners.

### Free and Open-Source Software Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

### Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

Synopsys, Inc.  
[www.synopsys.com](http://www.synopsys.com)